

The presentation follows the three thesis subsections in order, including every panel of Figures 3.20–3.26. Figures with several panels are followed by enlarged views.

Directional structure, terrain and equipment

Version: 14/09/2026

Thesis Sections 3.5.1, 3.5.2 and 3.5.3

Thesis §3.5.1

East–north balance in the retained archive

$$\rho_{EN}(t) = \frac{\langle E_f(t)^2 \rangle_f}{\langle N_f(t)^2 \rangle_f}, \quad \langle E_f^2 \rangle_f = \text{Var}_f(E_f) + \langle E_f \rangle_f^2.$$

$E_f(t)$ and $N_f(t)$ are positions relative to launch. The clock is time since launch; each time uses the flights that still contribute paired observations.

$\rho_{EN} > 1$: larger east second moment. $\rho_{EN} < 1$: larger north second moment.

The ratio combines mean displacement and spread. Equality is one necessary condition for isotropy; cross moments and the full angular distribution also matter.

Thesis Section 3.5.1.

The ratio compares motion along east and north

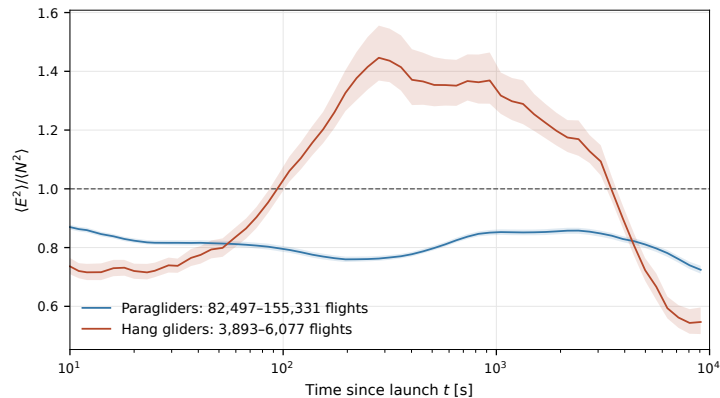
Discussion. How much of the imbalance comes from mean motion, and how much from spread around the mean?

Launch-referenced positions may retain reacquisition offsets across missing intervals. The subsequent PCA uses increments wholly inside segments; these avoid crossing gaps but do not independently validate their endpoints.

The retained archive has unequal east and north moments

Discussion. Would a fixed population change the time dependence of either ratio?

The directional imbalance motivates studying east, north and radial displacement separately. Route choice, terrain, wind and selection can all contribute. The curve ranges are not limits of the uncertainty bands.



Curve ranges over
10–10,000 s:

Paragliders: 0.72–0.87.

Hang gliders: 0.54–1.45.

The paraglider curve stays
below one. The hang-glider
curve crosses it.

Thesis Fig. 3.20, complete figure. Numbers are extrema of the observed curves; shading is the bootstrap range.

A site–day groups flights with the same recorded take-off site and date. Flights in one group may share weather and route conditions.

1. Resample whole groups with replacement, keeping their flights together.
2. Recompute the paired east/north moment ratio at every supported time.
3. Repeat 200 times; retain the 10th and 90th percentiles.

The shading contains the central 80% of bootstrap estimates at each time. Plots require 30 flights and 10 contributing groups.

Thesis Fig. 3.20 and the uncertainty convention used in Figs. 3.24–3.26.

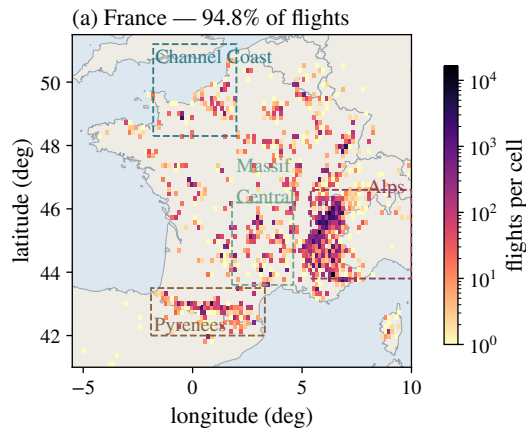
The bands resample whole take-off sites and days

Discussion. How much dependence remains across nearby sites, successive days or repeated pilots?

These are pointwise ranges, not simultaneous bands or the spread of individual flights. Missing site/date keys become singleton groups. Resampling preserves observed within-group dependence; it assumes groups can be independently resampled. It does not match equipment groups on weather.

Thesis §3.5.2

Regional principal axes



Retained paragliders and hang gliders are pooled. The map window contains 94.8% of their recorded starting locations.

Colour gives flights per $0.15^\circ \times 0.15^\circ$ cell, on a logarithmic scale.

Dashed boxes assign regional membership from the starting location: Alps, Pyrenees, Massif Central and Channel Coast.

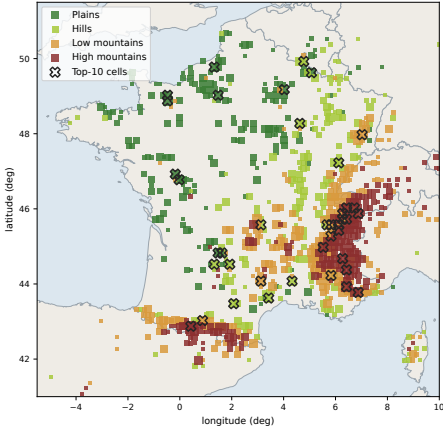
The PCA below compares Alps, Pyrenees and Channel Coast. Routes may leave their starting box.

France: launch density and regional selection boxes

Discussion. Would membership based on the full route change the regional comparison?

The colour scale counts retained flights, not distinct sites or time spent in a cell. The percentage refers to the plotted window, which includes neighbouring countries, not an administrative definition of France. Box bounds (longitude; latitude, degrees) are Alps: 5.4–10.0; 43.8–46.6, Pyrenees: -1.9–3.3; 42.0–43.5, Massif Central: 1.8–4.6; 43.6–46.2, and Channel Coast: -1.8–2.0; 48.3–51.2. The map pools both disciplines, whereas the regional PCA panels shown next concern paragliders. The coastal box includes cliffs and valleys, so its name does not establish low relief along a flight.

Chapter 2 geographic-coverage map, France panel. Boxes are latitude/longitude selections; they do not trace terrain boundaries or route exposure.



First raw IGC altitude z_0 : GNSS when usable, with pressure fallback for this map.

Map label	Recorded altitude
Plains	$z_0 < 300$ m
Hills	$300 \leq z_0 < 800$ m
Low mountains	$800 \leq z_0 < 1500$ m
High mountains	$z_0 \geq 1500$ m

Each $0.15^\circ \times 0.15^\circ$ cell: colour shows the most frequent altitude band; marker area grows with log flight count.

Crosses mark the ten busiest cells in each band. Both disciplines are pooled.

These labels describe starting altitude; they do not measure slope or local relief.

Chapter 2 starting-altitude map. Raw first-fix proxy with uncertain vertical datum; the independent ETOPO terrain map is introduced later.

France: recorded starting-altitude bands

Discussion. Would verified launch elevations and local relief change the altitude classification?

The map uses the first valid raw B record of each retained flight, before trajectory cleaning or trimming. GNSS is preferred when finite, nonzero and inside the configured plausibility range; otherwise a usable pressure altitude is used. Flights without a usable first altitude do not enter the coloured cells. The pressure fallback is specific to this exploratory map: the cleaned trajectory uses GNSS altitude. Marker area is $6 + 22 \log_{10}(n + 1)$, where n counts classified flights in the cell, and the colour is the modal band rather than the mean altitude. A first fix may precede take-off or occur after flight has begun. The four labels are altitude bins, not terrain-model classes; low starting altitude can coexist with cliffs or steep slopes.

$$\mathbf{X}_f(t, \tau) = \mathbf{r}_f(t + \tau) - \mathbf{r}_f(t), \quad \boldsymbol{\mu}_g = \langle \mathbf{X} \rangle_g,$$

$$\Sigma_g(\tau) = \left\langle (\mathbf{X} - \boldsymbol{\mu}_g)(\mathbf{X} - \boldsymbol{\mu}_g)^\top \right\rangle_g = U_g \text{diag}(\lambda_1, \lambda_2) U_g^\top.$$

The leading eigenvector gives the axis of greatest spread; λ_1 / λ_2 measures covariance anisotropy.

Use every supported increment in the regional archive on the 10-s grid. Origins are spaced by the lag within each segment. Flights with more origins receive more weight.

Thesis Section 3.5.2. Lags: 10, 100, 1000 and 10,000 s.

PCA measures spread around the mean displacement

Discussion. Would equal flight weights or fixed membership change the regional axes?

No increment crosses a segment boundary. The available population can shrink with lag. Eight flights is the display minimum, not a precision guarantee. Angles run counterclockwise from east, modulo 180 degrees; they become poorly determined when the two eigenvalues are close.

The ellipses show shape and direction at four lags

The ellipses show shape and direction at four lags

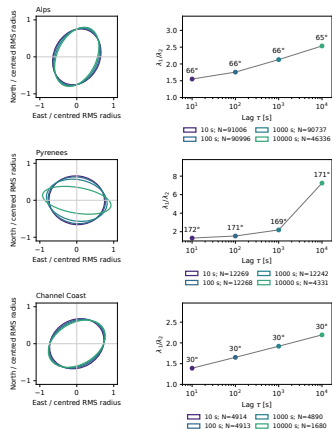
Normalising by one scalar preserves the eigenvalue ratio. This differs from whitening, which separately normalises the principal directions. The following slides enlarge the three original rows.

Both coordinates are divided by the same centred RMS radius:

$$s_g(\tau) = \sqrt{\text{tr} \Sigma_g(\tau)}.$$

$$\mathbf{x}^* = \frac{\mathbf{x} - \mu_g}{s_g(\tau)}.$$

Each region and lag has its own factor. Shape and angle are preserved; absolute displacement amplitude is removed.

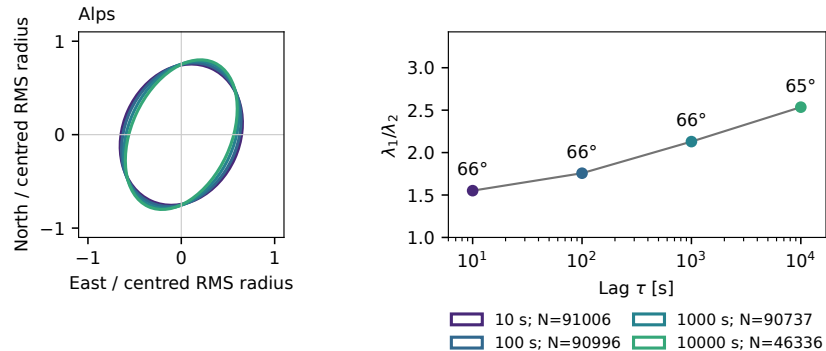


Thesis Fig. 3.21, all three rows. Right: eigenvalue ratio; labels give axis angles.

Alps: the principal axes change with lag

Discussion. How much of the change follows route geometry, and how much follows the changing population?

The plotted axis is modulo 180 degrees. A nearly circular ellipse does not determine a precise preferred axis. Coastal take-offs do not provide an isotropic reference by definition.



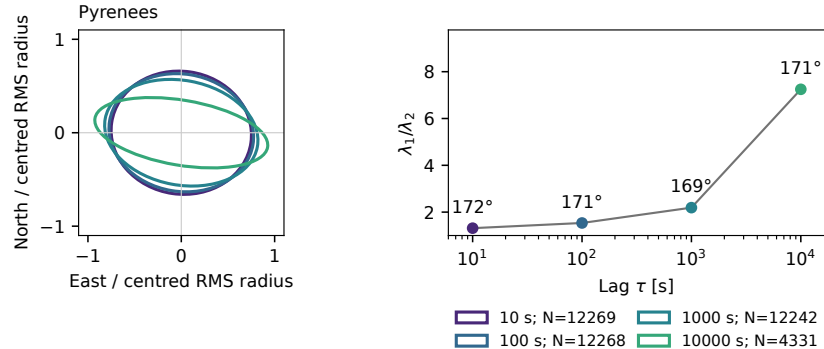
At 1000 s: $\lambda_1/\lambda_2 = 2.13$, axis 65.7° ; 90,737 flights.

Thesis Fig. 3.21, original Alps row. Each ellipse uses its own centred RMS radius.

Pyrenees: the principal axes change with lag

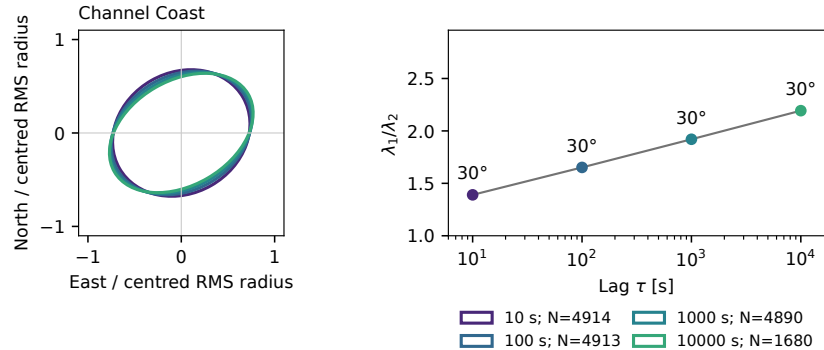
Discussion. How much of the change follows route geometry, and how much follows the changing population?

The plotted axis is modulo 180 degrees. A nearly circular ellipse does not determine a precise preferred axis. Coastal take-offs do not provide an isotropic reference by definition.



The axis approaches east–west and the eigenvalue ratio rises at long lag, while the contributing population falls.

Thesis Fig. 3.21, original Pyrenees row. Each ellipse uses its own centred RMS radius.



At 1000 s: $\lambda_1/\lambda_2 = 1.92$, axis 29.7° ; 4,890 flights.

Thesis Fig. 3.21, original Channel Coast row. Each ellipse uses its own centred RMS radius.

Channel Coast: the principal axes change with lag

Discussion. How much of the change follows route geometry, and how much follows the changing population?

The plotted axis is modulo 180 degrees. A nearly circular ellipse does not determine a precise preferred axis. Coastal take-offs do not provide an isotropic reference by definition.

λ_1/λ_2	10 s	100 s	1000 s	10,000 s
Alps	1.55	1.76	2.13	2.54
Pyrenees	1.31	1.54	2.19	7.25
Channel Coast	1.39	1.65	1.92	2.19

The centred displacement cloud becomes more elongated over longer intervals. Both principal variances may still grow.

Physical interpretation: longer increments can reveal route constraints and persistent wind contributions that are less visible over a short part of a flight.

Here τ is increment duration. The four measured lags do not establish a positive derivative at every scale; the contributing flights also change.

Thesis Section 3.5.2 and Fig. 3.21. Regional paraglider PCA; the environmental mechanisms remain hypotheses.

Directional constraints become more visible at longer lags

Discussion. Does the increase persist with fixed flights and origins, and does its scale follow terrain geometry or wind persistence?

This develops the interpretation that a pilot needs time to explore a corridor or accumulate persistent forcing before its signature becomes prominent in net displacement. The ratio compares proportional growth: $d \log(\lambda_1/\lambda_2)/d\tau = \lambda'_1/\lambda_1 - \lambda'_2/\lambda_2$ wherever differentiable. It is not a force measurement or the time since launch. The short-lag ellipses are already anisotropic. Declining support and changing route mixtures are alternative contributors. Values are rounded from the saved full regional PCA report.

τ_{terrain} is an illustrative time to explore a distance L across a valley or flight corridor; it has not been measured here.

When an increment spans such a distance, terrain or the edge of usable lift may require a turn. Across-corridor displacement is limited, while motion along the corridor can continue.

$$\tau_{\text{terrain}} \sim \frac{L}{v_{\perp}} \quad (\text{persistent crossing}), \quad \tau_{\text{terrain}} \sim \frac{L^2}{D_{\perp}} \quad (\text{diffusive exploration}).$$

$v_{\perp}$: transverse speed; $D_{\perp}$: transverse diffusion coefficient.

The two estimates assume different types of motion. Pilots may anticipate terrain, and local lift can affect headings at shorter lags.

Thesis Section 3.5.2, environmental interpretation. Actual terrain constraints depend on altitude, lift and route choice.

Terrain constraints appear as a route explores their scale

Discussion. Does the scale of the anisotropy increase track corridor width or distance to terrain along the actual route?

The idea of feeling a boundary means that an increment spans enough motion for terrain geometry to constrain its endpoint. Mountains are not reflecting walls, and pilots need not reach a physical edge: anticipated turns and the distribution of usable lift can guide them earlier. The relevant speed or diffusivity is transverse to the corridor. This interpretation allows a stronger signature at long lag without assuming zero terrain influence at short lag. Both sides of a curved corridor and mixtures of route directions complicate a single regional PCA axis.

Wind displacement accumulates as $\Delta \mathbf{r}_w(t, \tau) = \int_0^\tau \mathbf{w}_h(t+s) ds$. Direction changes can cause cancellation; persistence allows coherent accumulation.

Illustration: an isotropic background with covariance $2D\tau I_2$, plus an independent wind along $\mathbf{e}$, varying across observations with variance σ_w^2 and constant within each increment:

$$\Sigma(\tau) = 2D\tau I_2 + \sigma_w^2 \tau^2 \mathbf{e}\mathbf{e}^\top, \quad \frac{\lambda_1}{\lambda_2} = 1 + \frac{\sigma_w^2 \tau}{2D}.$$

A weaker contribution needs a longer interval to compete with background spread, provided it stays coherent that long.

A constant wind shared by all observations shifts only the mean and cancels from this centred PCA.

Thesis Section 3.5.2, illustrative wind model (not fitted). Temporal-correlation relation: Taylor (1922).

A weak persistent wind contribution can emerge with lag

Discussion. Is wind coherent over the inferred crossover, and are its directions aligned across the contributing flights?

The illustrative crossover is of order $2D/\sigma_w^2$, where σ_w^2 is the variance of the scalar wind component. It must lie within the wind-coherence regime. A nonzero common mean wind is removed by centring; the covariance term comes from variability across the ensemble. Temporal and spatial wind changes or route responses can also contribute. For a stationary scalar wind component, integrated displacement variance is $2 \int_0^\tau (\tau-s) C_w(s) ds$. Nearly constant random wind gives quadratic growth over short coherent intervals; finite integrable memory generally gives linear growth at long intervals. Longer observation does not create coherence. The real air-relative motion can correlate with wind, and persistent winds with isotropically distributed headings need not create an ensemble axis.

A rotation into the principal frame makes the components uncorrelated at the fitted lag and preserves radial lengths:

$$|U_g^\top \mathbf{X}| = |\mathbf{X}|.$$

Whitening with a reference covariance changes lengths:

$$\mathbf{Z}(\tau) = \Sigma_g(\tau_0)^{-1/2}[\mathbf{X}(\tau) - \boldsymbol{\mu}_g(\tau)].$$

Its covariance is the identity at τ_0 . At other lags, that remains a property to test. Whitening each lag separately would enforce equal principal variances in every ellipse.

Thesis Section 3.5.2. Neither uncorrelated components nor an isotropic covariance establish a rotationally invariant law.

Rotation and whitening change different properties

Discussion. Does one transformation fitted at a reference lag also describe other lags and flights?

Rotation cannot improve radial collapse because it preserves every radius. Whitening is discussed as a control; the ellipses in Fig. 3.21 use scalar trace normalisation.

$$W_\tau = \Sigma_g(\tau)^{-1/2}, \quad \mathbf{Z}_\tau = W_\tau[\mathbf{X}_\tau - \boldsymbol{\mu}_g(\tau)].$$

$$\text{Cov}(\mathbf{Z}_\tau) = W_\tau \Sigma_g(\tau) W_\tau^\top = I_2.$$

At each lag, every unit-vector projection has variance one. The covariance ellipse is a circle by construction.

This requires a positive-definite covariance. With estimated matrices, the identity holds on the same weighted observations used for fitting.

Angular structure beyond covariance and dependence between times remain to be tested.

Thesis Section 3.5.2. Whitening identity: Kessy, Lewin and Strimmer (2018), doi:10.1080/00031305.2016.1277159.

Whitening each lag makes each increment covariance isotropic

Discussion. Which angular or temporal differences remain after removing each lag's covariance shape?

Yes, each centred increment distribution becomes isotropic at second order. This statement concerns one lag at a time and the fitted population. It neither ensures the same property on held-out flights nor establishes a rotationally invariant full distribution. A Gaussian vector with zero mean and identity covariance is isotropic, but Gaussianity has not been established for these pooled increments. Lag-specific whitening removes the eigenvalue-ratio trend by definition and therefore cannot be used as evidence that environmental effects have been physically removed.

1. Crop NOAA ETOPO 2022 to the Alpine and Pyrenean take-off boxes.
2. Keep every fourth grid cell in both directions: one-arc-minute spacing.
3. Select cells at or above 1000 m; give higher cells no extra weight.
4. Project their centres to local WGS84 east/north coordinates at the box centre.
5. Weight each cell by $\cos(\text{latitude})$ to approximate its area.
6. Compute the weighted centre and spatial covariance.
7. Take its leading eigenvector as the mean highland axis.

Thesis Section 3.5.2, Independent terrain orientation. Zero projection height; angles counterclockwise from east, modulo 180° .

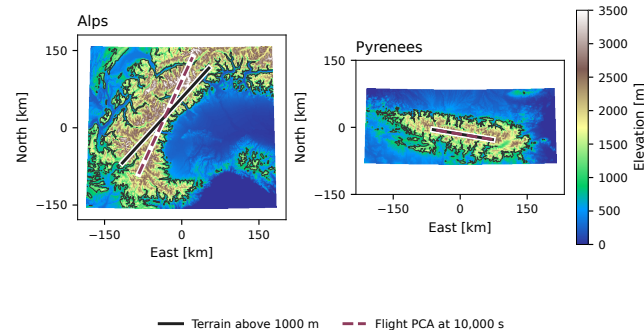
The terrain axis comes from an independent elevation map

The selected cells form a binary highland footprint. The construction estimates its elongation, rather than tracing a crest. No flight enters the terrain calculation, and the threshold was not fitted to the flight axis.

Terrain and flight axes align more closely in the Pyrenees

Discussion. Would terrain axes along the observed tracks explain the Alpine offset?

The Pyrenean comparison supports a geographical association. The Alpine chain is curved and truncated by the box. Wind, route choice and the long-flight population may also contribute. These maps use a single tangent frame per region, while flight increments use their trajectory frames and may leave the box.



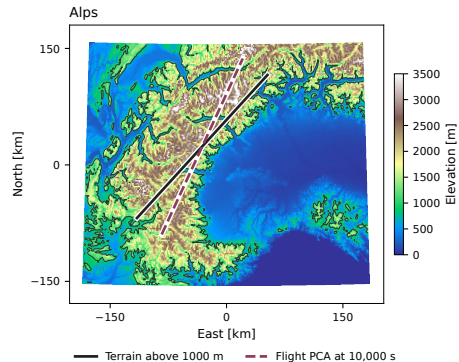
Pyrenees: separation below 1° . Alps: 17.6° , with the flight axis more northward.

Thesis Fig. 3.22, both maps. Solid dark: terrain axis; dashed colour: flight PCA at 10,000 s. Line lengths carry no physical information.

Alps: shared sector, with an approximately 18-degree offset

Discussion. How much of the offset reflects Alpine curvature, box truncation or route selection?

Both orientations occupy the northeast-southwest sector, but the flight covariance is more northward. A simple claim that flights run parallel to one Alpine axis misses the approximately eighteen-degree difference. The terrain angle spans 42.1 to 47.9 degrees over the three thresholds, so this sensitivity does not eliminate the offset. Wind and long-flight selection remain possible contributors; the figure neither isolates a terrain effect nor tests pilot skill.



Terrain: 47.6° .

Flight PCA: 65.2° .

Separation: 17.6° .

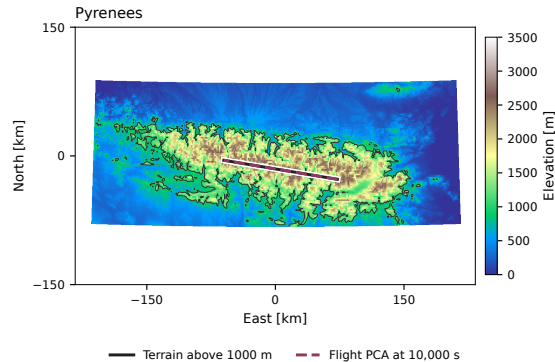
46,336 flights support 10,000 s.

Black: highland-footprint axis. Dashed colour: current flight PCA, shifted to the same centre. Line lengths have no physical meaning.

Pyrenees: the long-lag spread follows the highland axis

Discussion. Does the alignment persist for fixed flights and for terrain sampled along their routes?

The near-parallel axes support a geographical association between displacement orientation and the range. They do not establish the causal contribution of terrain. Winds can align with the same range and pilots can select routes along it. The sensitivity span is not a sampling confidence interval. The comparison is made within the exact Pyrenean take-off box, but flight increments can extend beyond it.



Terrain: 170.5° .

Flight PCA: 170.5° .

Separation: less than 1° .

4,331 flights support 10,000 s.

ETOPO footprint above 1000 m; threshold sensitivity: $169.7\text{--}175.4^\circ$. Decimal agreement does not imply this physical precision.

Region	Terrain axis	Flight axis	Separation
Alps	47.6°	65.2°	17.6°
Pyrenees	170.5°	170.5°	<1°

Thresholds of 500, 1000 and 1500 m give:

Alps: 42.1°–47.9°, Pyrenees: 169.7°–175.4°.

These spans measure sensitivity to the selected highland area. They are not confidence intervals. Decimal agreement does not imply that the physical mountain direction is known to that precision.

Thesis Section 3.5.2, commentary following Fig. 3.22.

The terrain comparison depends on the footprint definition

Discussion. Does the comparison change with the box boundary or a local valley direction?

The main PCA has 46336 Alpine and 4331 Pyrenean flights at this lag. Neither map measures the terrain exposure of individual flights.

Same population as the 10,000-s PCA

- 1,680 paraglider flights.
- 2,013 nonoverlapping windows, wholly inside retained segments.
- Mean cleaned GNSS altitude along each window; equal window weights.

Trapezoidal integration on the same 10-s segment grids. The target describes the supported long-lag windows; it is not take-off altitude or height above the terrain.

Mean: 1,006 m

Equal flight weights: 980 m.

Middle half of window means:
854 m–1,160 m.

Approximate altitude above sea level;
recorder datums are not harmonised.

Channel Coast: the reference height follows the PCA windows

Discussion. Is one representative height sufficient, or should wind be matched to the altitude and location of each flight window?

The mean is 1006.262 metres across 2013 exact PCA windows from 1680 flights, with the same equal-window weighting as the covariance. Each window altitude is a trapezoidal time average of cleaned GNSS z interpolated onto the existing 10-second grid. No gap is bridged. Equal weight per contributing flight gives 980.260 metres; the window-mean quartiles are 853.932 and 1159.549 metres. Short flights do not define this target because they cannot contribute at 10000 seconds. The raw GNSS vertical datum is not harmonised, so MSL is an approximation; these numbers do not imply metre-level physical accuracy. The quartile range describes heterogeneity, not uncertainty in the mean. Matching the target height does not yet match flight dates, instantaneous positions or local atmospheric conditions.

2016–2025, hourly; nine cells in the coastal box; 1000/925/850/700 hPa.

$$H = \Phi / g_0, \quad h = \frac{R_{\oplus} H}{R_{\oplus} - H}, \quad \theta_w = \text{atan2}(\langle v(h) \rangle, \langle u(h) \rangle).$$

Convert geopotential to each hour's level heights. Interpolate **E/N wind components** to 1,006 m, between above-ground bounds.

Main comparison: all year, all hours vs. all year, 09–17 UTC.

April–September is a separate seasonal check. Winds at 10/100 m AGL provide lower-level controls.

ECMWF/Copernicus ERA5 pressure levels, accessed through a pinned snapshot of Earthmover's public cloud-optimised ERA5 copy on AWS. Earthmover distributes this lossless copy; it is not a separate reanalysis. Complete hourly extract: 32.22 MB compressed, preserved in the repository and on SSD.

ERA5 wind is interpolated to the measured flight altitude

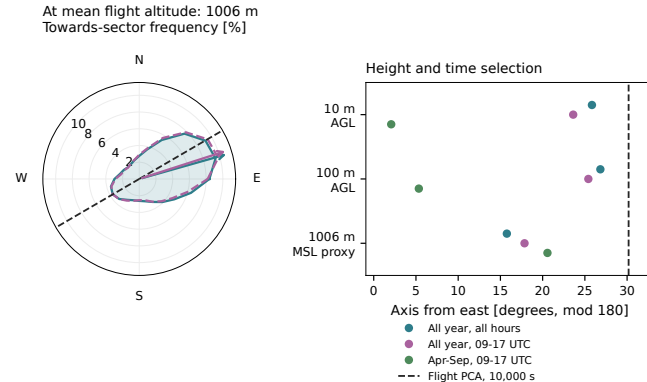
Discussion. How much of the surface-to-flight-height contrast reflects vertical shear, season and the flight conditions represented?

Use the same nine grid cells as the surface analysis: latitudes 48.75, 49.75, 50.75 degrees and longitudes minus 1.25, zero, 1.25 degrees. Geopotential height is Phi divided by 9.80665; geometric height uses spherical Earth radius 6371000 metres. A pressure level changes altitude from hour to hour. Linear interpolation acts on eastward and northward velocities, never direction angles. All 789048 cell-hour profiles bracket the main target above the model surface. Average components over time and with cosine-latitude cell weights, then reduce the towards-angle modulo 180 degrees for comparison with PCA. The annual all-hour selection has 87672 hours per cell; 09 through 17 UTC inclusive has 32877. The two warm-season selections have 43920 and 16470. UTC is fixed, without a daylight-saving correction. Source values, timestamps, coordinates, pressure levels, geopotential, surface geopotential and hashes are preserved losslessly for offline reuse.

At flight height, both annual references remain offset from PCA

Discussion. Why is the seasonal contrast larger near the surface?

At the mean flight altitude the all-year all-hour axis is 15.8 degrees and the 09–17 UTC axis is 17.9 degrees, against 30.2 degrees for flight PCA. The corresponding separations are 14.4 and 12.3 degrees. At 10/100 metres AGL the annual axes lie closer to PCA, but their warm-daytime references rotate further away. At flight altitude the warm-daytime axis is 20.6 degrees, giving a smaller 9.6-degree separation. The altitude-matched comparison therefore has a less close annual alignment and a smaller time-selection sensitivity than the surface comparison. These are descriptive contrasts, not formal tests of different angles. The rose uses weighted 15-degree sectors, excludes speeds below 0.5 m/s only from frequencies, and retains all hours in vector means. Teal is all-year all hours; plum is all-year daytime. Green appears only in the right-hand seasonal check.



Annual gaps at flight height: 14.4° for all hours, 12.3° for 09–17 UTC.

Thesis Fig. 3.23, complete figure. Angles counterclockwise from east. The rose shows where air moves towards at 1,006 m; the black line is the 30.2° centred PCA axis.

At 1,006 m	Wind axis	PCA gap	ρ_w
All year, all hours	15.8°	14.4°	0.43
All year, 09–17 UTC	17.9°	12.3°	0.44
Apr–Sep, all hours	16.0°	14.2°	0.35
Apr–Sep, 09–17 UTC	20.6°	9.6°	0.37

At the quartile heights, annual axes span 14.3–17.2° (all hours) and 16.4–19.3° (daytime), on shared support.

$\rho_w = \|\langle \mathbf{w} \rangle\| / \langle \|\mathbf{w}\| \rangle$ measures cancellation. Height ranges are sensitivities, not confidence intervals. The main estimates retain all selected hours.

Height and time checks preserve a broad orientation resemblance

Discussion. Does the remaining offset reflect flight selection and route choices, or change when weather is matched to actual flights?

The annual all-hour and daytime resultant fractions are 0.43 and 0.44; warm-season values are 0.35 and 0.37. These do not give confidence in a direction or the fraction of observations near it. At the lower quartile height, 868 profiles at one cell have no above-ground pressure level below the target. For the height sensitivity only, exclude these 868 hours from all nine cells at every tested height, including the mean-height reference: 86804 common hours remain per cell. Apply the same seasonal and daytime masks afterward. On common support the annual quartile-height axes range from 14.3 to 17.2 degrees for all hours and from 16.4 to 19.3 degrees for daytime. Equal-flight mean altitude changes the matched reference by less than 0.3 degrees, and common-hour selection changes the main annual reference by less than 0.1 degrees. These checks retain the offset but do not calibrate receiver datums or establish significance.

Region	Conclusion supported by this comparison
Pyrenees	Close terrain–flight alignment; a geographical association worth testing along tracks.
Alps	Same broad sector, with an approximately 18-degree offset from the mean highland axis.
Channel Coast	Wind at mean flight altitude lies about 12–14 degrees from PCA in both annual time selections; seasonal checks retain a broad resemblance.

A common constant wind shifts the mean displacement by $\tau\mathbf{w}$; centring removes it. Variable winds and route responses can still shape covariance.

Independent regional references constrain an interpretation. They do not separate wind, terrain, route choice and long-flight selection.

What the environmental comparisons establish

Discussion. Match flight times, heights and routes to wind and local terrain before attributing the ellipse or equipment contrasts to a cause.

The PCA concerns centred displacement spread, whereas the wind reference concerns a mean velocity. Their orientation resemblance alone cannot show that mean wind causes covariance anisotropy. Constant additive drift cancels when displacement is centred at fixed lag. Variable drift between flights, time-varying forcing and route decisions can alter covariance; their contributions need matched observations. The Pyrenean agreement is the strongest geographical alignment here. The coastal comparison at mean flight altitude retains a broadly similar sector across the four time selections, with offsets of about 10 to 14 degrees. Surface winds exhibit a larger seasonal change, so their annual near alignment should not be transferred to flight altitude. Equipment class is not measured pilot expertise, and this comparison cannot establish resistance to wind.

MSL proxy: the recorded GNSS altitude is treated approximately as altitude above mean sea level; receiver datums have not been harmonised.

AGL: 10 and 100 m above the ERA5 model surface. These are lower-level wind controls, with a different vertical reference.

$$\rho_w = \frac{\|\langle \mathbf{w} \rangle\|}{\langle \|\mathbf{w}\| \rangle}.$$

$\rho_w = 1$ means no directional cancellation; opposite winds can give a small resultant even when their individual speeds are large. Annual values of 0.43 and 0.44 indicate substantial cancellation.

Thesis Section 3.5.2, coastal wind reference. ρ_w is not an angular confidence level.

Reading the height labels and the mean wind

Discussion. Would flight-matched wind directions differ from the climatological vector mean?

Interpolate and average east/north wind components, then calculate the angle. The climatology is not paired with each flight date or instantaneous altitude. The height sensitivity checks preserve the annual offset; they do not calibrate GNSS datums.

Thesis §3.5.3

Terrain, equipment and wind

Position, velocity and acceleration have different imbalances

Discussion. Which directional differences persist after comparing the same sites, dates and flight durations?

Each panel has its own vertical range. These plots fit no scaling exponent. A ratio close to one leaves cross moments and the angular law untested.

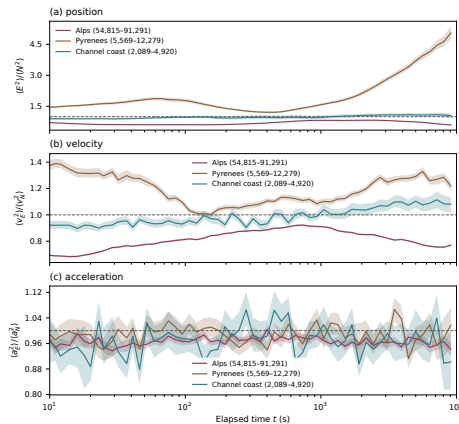
All three panels use raw east/north second moments at time since launch.

Median position / velocity ratios:

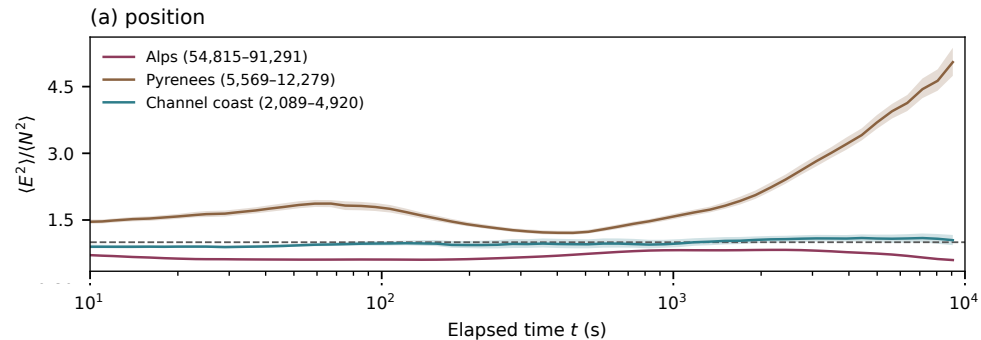
Alps: 0.67 / 0.81.

Pyrenees: 1.67 / 1.19.

The Pyrenean east–west preference is consistent with terrain influencing headings; wind and route selection remain possible contributors.



Thesis Fig. 3.24, all three panels, including acceleration. Shading: pointwise 10–90% site–day bootstrap ranges.

Regional position moments differ along east and north

The Pyrenean position ratio rises above five at late times. The Alpine curve stays below one; the coast lies closer to one.

Thesis Fig. 3.24(a), enlarged original panel.

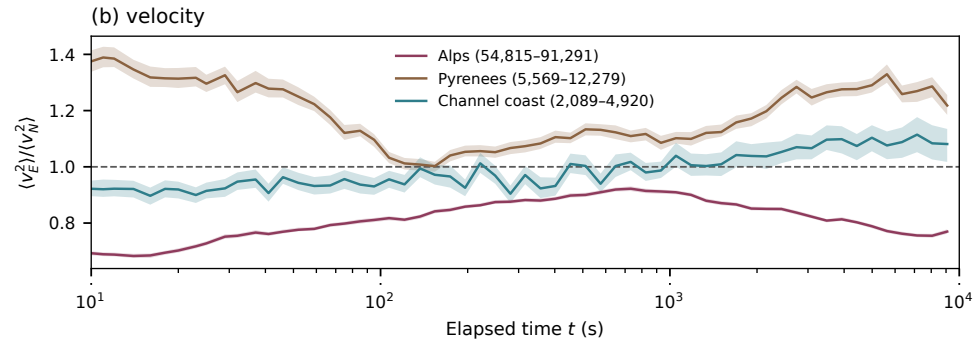
Discussion. How much imbalance reflects mean drift, route orientation or changing support?

The clock is time since estimated launch, not an increment lag. Position ratios approach five in the Pyrenees. Raw second moments include both covariance and the squared mean. Splits prevent cross-gap stencils but recorded endpoint offsets can still affect these launch-referenced coordinates.

Regional velocity moments show a related directional contrast

Discussion. Which position and velocity contrasts persist under matched conditions?

The regional figure pools paired east/north observations at each elapsed time and uses site-day pointwise bands. Terrain, weather, routes and the disappearing short-flight population can all contribute. Coordinate-moment balance is weaker than full isotropy.



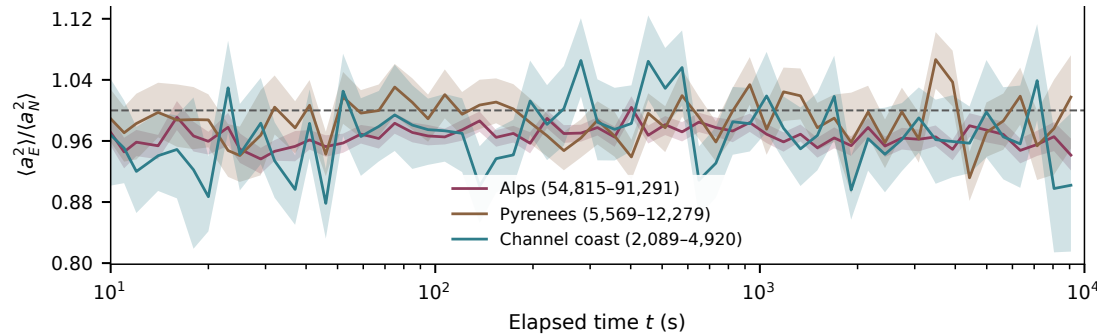
East–west preference is also visible in Pyrenean velocity moments. The relative spread varies less than the position ratio.

Thesis Fig. 3.24(b), enlarged original panel.

Regional acceleration ratios are closer to unity

Discussion. How much of this balance comes from manoeuvres, route mixing or wind gradients?

The acceleration ratio is approximately one in the three groups, but equal east and north moments do not establish isotropy of the acceleration distribution. Differentiation only removes a constant additive wind.



Acceleration has more balanced coordinate moments than the large position contrasts.
Spatially varying wind can still contribute.

Thesis Fig. 3.24(c), original panel. This is a ratio of raw second moments.

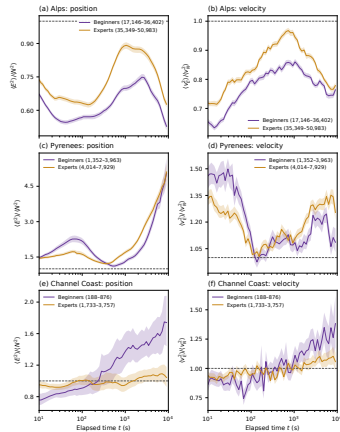
Equipment contrasts differ between the three regions

Certification describes tested flight behaviour and associated pilot requirements. It is not itself an aerodynamic performance ranking. An experienced pilot may fly an A/B wing.

Violet: EN A/B.
Ochre: EN C/D/CCC.

The legend uses these equipment groups as beginners/experts proxies.

Actual pilot skill is unmeasured.
Tandem, non-certified and unclassified records are excluded.

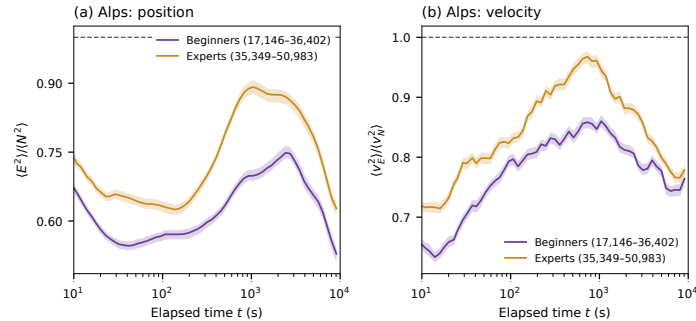


Thesis Fig. 3.25, all six panels. The next slides enlarge each original row.

Alps: C/D/CCC has smaller coordinate-moment imbalance

Discussion. Does the ordering persist after matching site, day, task and duration?

The fresh Alpine position and velocity ratios stay closer to unity for C/D/CCC over the supported grid. This is a descriptive ensemble observation. It does not establish individual isotropy, greater skill or a causal equipment effect. Violet: A/B; ochre: C/D/CCC. Beginners and experts in the retained figure legend denote equipment proxies. Bands are pointwise 10–90-percent site-day bootstrap ranges, not intervals for the group difference.



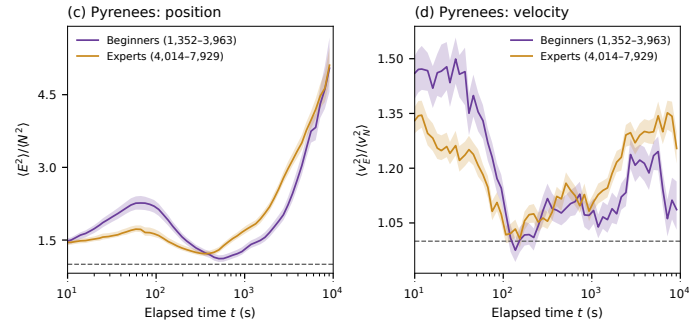
C/D/CCC stays closer to unity over the supported times. Median position ratio: 0.70, versus 0.61 for A/B.

Thesis Fig. 3.25, Alpine row.

Pyrenees: the ordering of the two groups reverses

Discussion. What changes between the early and later portions of the curves?

The Pyrenean curves do not support a uniform expert-proxy advantage. At late times both position ratios exceed five, while the velocity ordering also changes. Adjacent plotted times are correlated observations, not independent tests. Violet: A/B; ochre: C/D/CCC. Beginners and experts in the retained figure legend denote equipment proxies. Bands are pointwise 10–90-percent site-day bootstrap ranges, not intervals for the group difference.



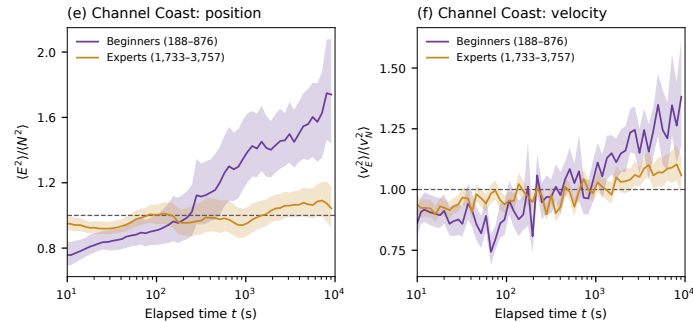
Neither group remains uniformly closer to one. Near 9086 s, position ratios are about 5.05 for A/B and 5.11 for C/D/CCC.

Thesis Fig. 3.25, Pyrenean row.

Channel Coast: the late position contrast is pronounced

Discussion. Can matching conditions separate directional route choice from wind response?

Near 9086 seconds the position ratios are about 1.74 for A/B and 1.04 for C/D/CCC, supported by 188 and 1733 paired flights. These are raw ensemble second moments and the samples differ strongly. Coastal cliffs and valleys prevent a wind-only interpretation. Violet: A/B; ochre: C/D/CCC. Beginners and experts in the retained figure legend denote equipment proxies. Bands are pointwise 10–90-percent site-day bootstrap ranges, not intervals for the group difference.



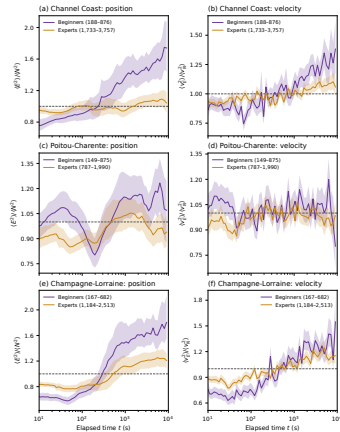
Near 9086 s: 1.74 for A/B and 1.04 for C/D/CCC, with 188 and 1733 contributing flights.

Thesis Fig. 3.25, coastal row.

Two additional regions extend the coastal comparison

Discussion. Does the coastal pattern persist in other regions and on matched flight conditions?

The boxes are broad geographic selections, not verified polygons of flat terrain. The following two slides enlarge the additional regions. The coastal row is the same comparison shown in Fig. 3.25.



Thesis Fig. 3.26, all six panels.

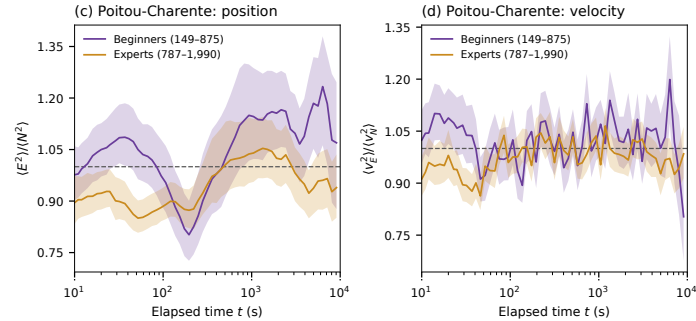
The coastal row is repeated to compare it directly with Poitou-Charente and Champagne-Lorraine.

These boxes classify take-offs. They do not measure the terrain crossed by each flight or ensure that the equipment groups experienced comparable weather.

Poitou-Charente: the group curves cross

Discussion. Does this low-relief launch box reproduce the coastal observation?

The crossing curves and broad overlapping bands do not support a general group ordering. These broad launch-location boxes are neither DEM-verified plains nor controls matched on weather. Violet: A/B; ochre: C/D/CCC. Beginners and experts in the retained figure legend denote equipment proxies. Bands are pointwise 10–90-percent site-day bootstrap ranges, not intervals for the group difference.



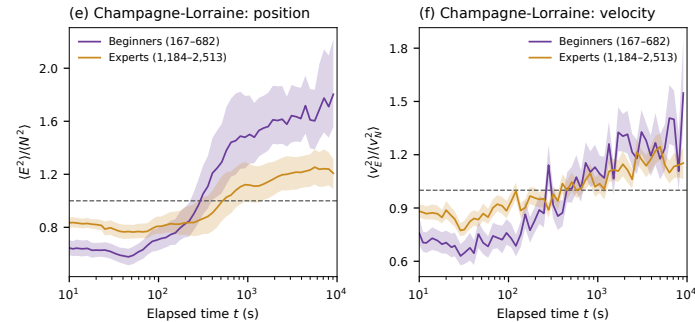
The curves cross and their bands overlap broadly. A consistent equipment ordering is not supported.

Thesis Fig. 3.26, middle row.

Champagne-Lorraine: a late contrast does not keep C/D/CCC at unity

Discussion. Which parts of the coastal hypothesis transfer to this separate region?

A later position contrast appears, but the C/D/CCC ratio also rises above one. Region, route mixture, weather and declining support must be separated before claiming the same physical cause. Violet: A/B; ochre: C/D/CCC. Beginners and experts in the retained figure legend denote equipment proxies. Bands are pointwise 10–90-percent site-day bootstrap ranges, not intervals for the group difference.



A late position contrast appears, but the C/D/CCC ratio also rises above one.

Thesis Fig. 3.26, bottom row.

Hypothesis to discuss

Equipment and pilot choices may moderate directional constraints associated with wind and terrain. Similar mountain profiles may reflect shared route constraints.

Evidence limiting that interpretation

The Pyrenean ordering reverses; class is an experience proxy; coastal terrain is not absent; late support differs; and ensemble balance does not establish individual isotropy or performance.

The environmental interpretation remains a hypothesis

Discussion. What observation would distinguish this hypothesis from different sites, tasks, weather or route mixtures?

Preserve the proposed explanation as a useful hypothesis rather than silently removing it or presenting it as established. Similar curve shape does not by itself establish a common cause or an unavoidable terrain constraint. Wind and terrain can interact, so the coast/mountains split does not identify their separate effects.

$$\rho_{EN} = 1, \quad \Sigma = \begin{pmatrix} 1 & 0.9 \\ 0.9 & 1 \end{pmatrix}, \quad \lambda_1/\lambda_2 = 19.$$

This centred Gaussian has balanced coordinate moments but strongly unequal principal variances.

A pooled mixture of east-directed and north-directed routes can also balance the ensemble while each flight remains directional.

Equal east and north moments do not establish isotropy

Discussion. Are we observing mean drift, centred directional variability, or a mixture of individually oriented routes?

Full isotropy requires rotation invariance of the law. An isotropic covariance alone would still be only a second-order condition. Raw moments additionally contain the outer product of the mean with itself. An intentional downwind route can be successful despite being directional, so this ratio is not a direct performance score.

The Channel Coast includes cliffs and valleys, including the Pays de Caux. Wind and terrain may jointly orient lift and routes.

Equipment groups can differ in sites, weather, task and duration. Their bootstrap bands reflect one grouping structure; the groups have not been meteorologically matched.

Poitou-Charente and Champagne-Lorraine do not reproduce a uniform C/D/CCC advantage. The coastal observation alone does not establish a general flatland mechanism.

Thesis Section 3.5.3, Equipment, terrain and selection.

Coastal flights still encounter terrain and selection effects

Discussion. Does the contrast survive when both equipment groups are observed on the same site–days?

Similar mountain curve shapes can suggest shared constraints but cannot identify a common cause. Wind and terrain interact. Equipment, pilot choice and task remain entangled.

$$\mathbf{v}_{\text{ground}} = \mathbf{v}_{\text{air}} + \mathbf{w}_h.$$

For a wind field $\mathbf{w}_h(\mathbf{r}, z, t)$,

$$\mathbf{a}_{\text{ground}} = \frac{d\mathbf{v}_{\text{air}}}{dt} + \partial_t \mathbf{w}_h + (\mathbf{v}_{\text{ground}} \cdot \nabla_h) \mathbf{w}_h + \dot{z} \partial_z \mathbf{w}_h.$$

A constant additive wind disappears under differentiation. A steady wind field can still affect acceleration when the glider crosses horizontal or vertical gradients.

Thesis Section 3.5.3, Wind and ground velocity.

Ground velocity combines air-relative motion and wind

Discussion. Which wind contribution can be estimated independently along the tracks?

Ground velocity alone cannot separate still-air motion from wind. Mean route choice and net progress also affect the observed trajectories. Randomly rotating flights preserves their radii and provides an alignment control, without recovering air-relative motion.

1. Restrict comparisons to shared take-off sites and days, with comparable flight duration and task.
2. Separate mean ground drift, centred spread and individual route directions.
3. Match wind to flight times and heights, and terrain to the actual tracks.
4. Where possible, compare the same pilot using different wings.

Upwind progress, air-relative performance and choice of route are different outcomes. The equipment interpretation depends on which one is being tested.

Thesis Section 3.5.3, Discriminating comparisons.

Compare equipment groups under similar flight conditions

Discussion. Which comparison can most directly distinguish route selection from an equipment effect?

Thermal-centre drift may supply an independent wind estimate after segmentation, subject to assumptions about thermal motion. Site–day bootstrap estimates uncertainty; matching addresses comparability.